

COMM50 Postgraduate Study & Research Methods

Devising a Research Question or Hypothesis

1. Think about areas of computing that interest you and note down some key words or phrases associated with that area. Use these notes to try to form broad areas of research. These could be based upon articles you have read recently or other fields that you have studied. Try to come up with at four or five topics. For example, these topics could be things like: “*Computer Vision*”; “*Voice over IP*”; “*User Experience*”; “*Routing Protocols*”; “*3D Computer Graphics*”; and so on.
2. From the list of topics you have created, try to focus these into much more specific and defined topic areas. For example: “*Computer Vision: Facial Recognition*”; “*Computer Vision: Eye Tracking*”; “*Voice over IP: Encryption*”; “*Voice over IP: Quality of Service*”; etc. Try and come up with at least two of these specific topics. You might find it helpful to use a mind map approach, to keep track of your themes.
3. Now, working with your list of more specific topics and incorporating your existing knowledge of the field (although you might want to expand this by doing some brief literature searches), see if you can form two or three research questions about each. A good way to approach this is to try and think about problems or new application areas that could be addressed. You might also think of lines of enquiry that explore the “*how*”, “*why*” or “*what if*”.
4. Reflect upon your list of potential questions. Do they communicate your research idea clearly and concisely? You may want to apply the FINER* criteria to help you do this. Try to refine each of your questions to ensure that they satisfy these criteria.

FINER Research Question Criteria	
F	Feasible Can you do this? Is there enough time? Do you have access to the resources and samples required?
I	Interesting Who is the research about? Who will be impacted by the research?
N	Novel What is new or different about your approach? What will you seek to demonstrate in this research?
E	Ethical Is the research ethically allowable in your institution? Who could be at risk? Does the scope need adjusting to make it compliant?
R	Relevant Will this work expand what is already known about the topic? Does it have application?

5. Attempt to refine each of your research questions (which should be specific, but imprecise) into research hypotheses (which should be specific and precise). This should be focused, touch upon the key method of testing, and the result you *expect* to see (hence you are producing an alternate hypothesis, rather than a null hypothesis). Take the following, for example:

Research Question	How would the choice of encryption key management protocols affect the quality of VoIP calls?
Hypothesis	<i>Measures of VoIP call quality (PSNR and SDG) will change by altering the Secure Real-time Transport Protocol (SRTP) key management protocol.</i>

Over the page, you can see an example of how each of these tasks might result in a mind map of research ideas. The example is small and is limited to the VoIP topic after the second level (sub-topic); yours will have more branches and nodes under each.

* Hulley S, Cummings S, Browner W, et al. *Designing clinical research*. 3rd ed. Philadelphia (PA): Lippincott Williams and Wilkins; 2007.

